1. Azure CLI on windows

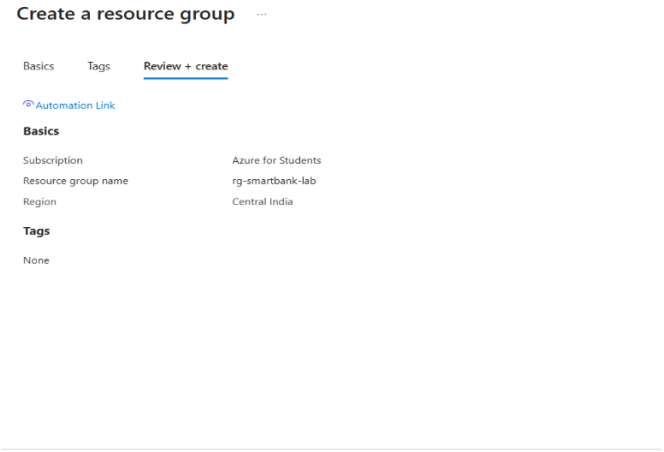
```
C:\Users\PAVILION>az

     /\
    /  \    _____   _ _ ___ _
   / /\ \  |_  / | | | \'__/ _\
  / ____ \  / /| |_| | | |  __/
 /_/    \_\/___|\__,_|_|  \___|


Welcome to the cool new Azure CLI!

Use `az --version` to display the current version.
Here are the base commands:


Group
    az

Subgroups:
    account                 : Manage Azure subscription information.
    acr                     : Manage private registries with Azure Container Registries.
    ad                      : Manage Microsoft Entra ID (formerly known as Azure Active Directory,
                              Azure AD, AAD) entities needed for Azure role-based access control
                              (Azure RBAC) through Microsoft Graph API.
    advisor                 : Manage Azure Advisor.
    afd                     : Manage Azure Front Door Standard/Premium.
    aks                     : Azure Kubernetes Service.
```

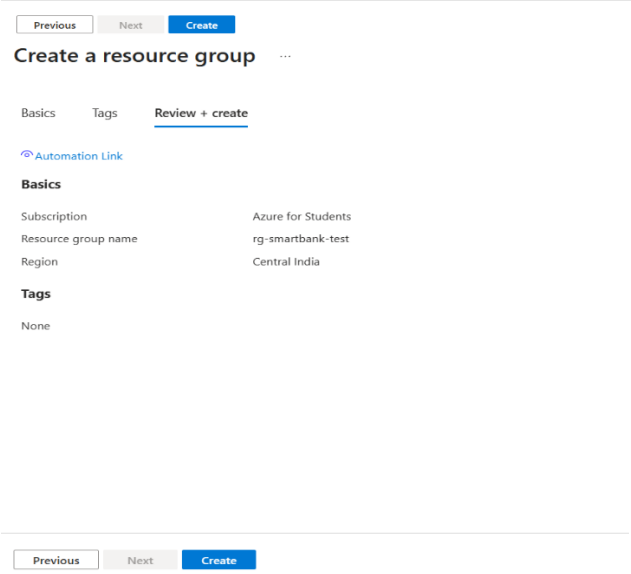
```
C:\Users\PAVILION>az login
Select the account you want to log in with. For more information on login with Azure CLI, see https://go.microsoft.com/f
wlink/?linkid=2271136

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No     Subscription name    Subscription ID                       Tenant
-----  -------------------  ------------------------------------  -----------------
[1] *  Azure for Students   d780980c-d1c6-4cf2-9353-c5838b3e1627  Default Directory

The default is marked with an *; the default tenant is 'Default Directory' and subscription is 'Azure for Students' (d78
0980c-d1c6-4cf2-9353-c5838b3e1627).

Select a subscription and tenant (Type a number or Enter for no changes):

Tenant: Default Directory
Subscription: Azure for Students (d780980c-d1c6-4cf2-9353-c5838b3e1627)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about i
t and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subsc
riptions by default.
```

```
C:\Users\PAVILION>az account show
{
  "environmentName": "AzureCloud",
  "homeTenantId": "42306a95-e3b0-4f49-90ce-b69da8a59d9c",
  "id": "d780980c-d1c6-4cf2-9353-c5838b3e1627",
  "isDefault": true,
  "managedByTenants": [],
  "name": "Azure for Students",
  "state": "Enabled",
  "tenantDefaultDomain": "ishanbansal2123gmail.onmicrosoft.com",
  "tenantDisplayName": "Default Directory",
  "tenantId": "42306a95-e3b0-4f49-90ce-b69da8a59d9c",
  "user": {
    "name": "ishanbansal2123@gmail.com",
    "type": "user"
  }
}
```

```
C:\Users\PAVILION>az group create --name rg-demo --location centralindia
{
  "id": "/subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-demo",
  "location": "centralindia",
  "managedBy": null,
  "name": "rg-demo",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}

C:\Users\PAVILION>az group create --name rg-demo --location centralindia
{
  "id": "/subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-demo",
  "location": "centralindia",
  "managedBy": null,
  "name": "rg-demo",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}

C:\Users\PAVILION>az group list --output table
Name                                                Location      Status
--------------------------------------------------  ------------  ---------
VisualStudioOnline-BD0E2BF97AA949AEA8BAEB0A4622533C  southindia    Succeeded
rg-demo                                             centralindia  Succeeded

C:\Users\PAVILION>az group delete --name rg-demo --yes --no-wait

C:\Users\PAVILION>
```

2. Lab1 Azure Portal and Resource Group





3. Lab2 Azure Cloud Shell + CLI

```
ishaan [ ~ ]$ az group create --name rg-cli-demo2 --location centralindia
{
  "id": "/subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-cli-demo2",
  "location": "centralindia",
  "managedBy": null,
  "name": "rg-cli-demo2",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
```

```
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

Storage fileshare subscription d780980c-d1c6-4cf2-9353-c5838b3e1627 is not registered to Microsoft.Cl
isterCloudShell" to register. In future, unregistered subscriptions will have restricted access to Cl

ishaan [ ~ ]$ az group create --name rg-smartbank-cli --location centralindia
{
  "id": "/subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-smartbank-cli",
  "location": "centralindia",
  "managedBy": null,
  "name": "rg-smartbank-cli",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
```

```
ishaan [ ~ ]$ az group list --output table
Name                                                Location      Status
--------------------------------------------------  ------------  ---------
VisualStudioOnline-BD0E2BF97AA949AEA8BAEB0A4622533C  southindia    Succeeded
rg-smartbank-lab                                    centralindia  Succeeded
rg-smartbank-test                                   centralindia  Succeeded
cloud-shell-storage-centralindia                    centralindia  Succeeded
rg-smartbank-cli                                    centralindia  Succeeded
rg-cli-demo2                                        centralindia  Succeeded
ishaan [ ~ ]$
```

4. Lab3 Azure PowerShell

```
PS /home/ishaan> New-AzResourceGroup -Name rg-demo -Location CentralIndia

ResourceGroupName : rg-demo
Location          : centralindia
ProvisioningState : Succeeded
Tags              :
ResourceId        : /subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-demo

PS /home/ishaan> New-AzResourceGroup -Name rg-smartbank-ps -Location CentralIndia

ResourceGroupName : rg-smartbank-ps
Location          : centralindia
ProvisioningState : Succeeded
Tags              :
ResourceId        : /subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-smartbank-ps

PS /home/ishaan> New-AzResourceGroup -Name rg-ps-demo2 -Location CentralIndia

ResourceGroupName : rg-ps-demo2
Location          : centralindia
ProvisioningState : Succeeded
Tags              :
ResourceId        : /subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-ps-demo2
```

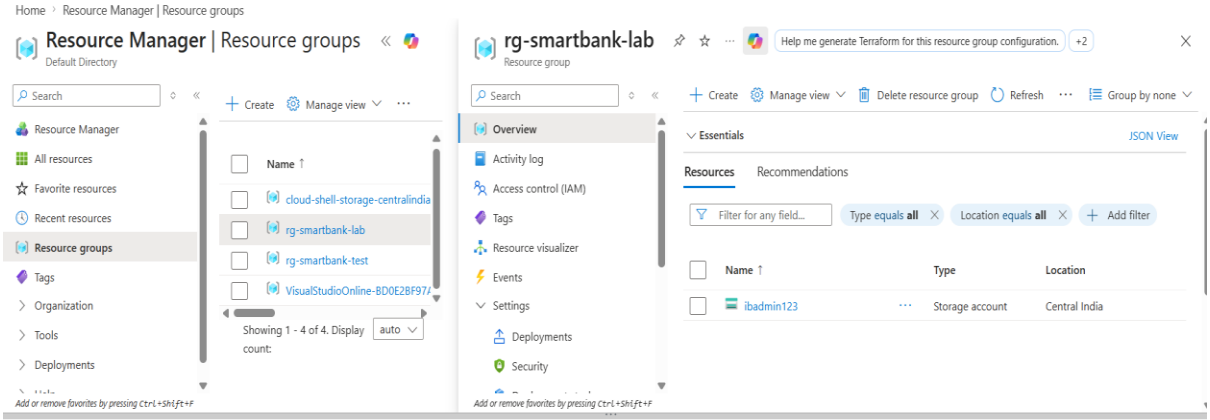
```
PS /home/ishaan> Get-AzResourceGroup

ResourceGroupName : VisualStudioOnline-BD0E2BF97AA949AEA8BAEB0A4622533C
Location          : southindia
ProvisioningState : Succeeded
Tags              :
ResourceId        : /subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/VisualStudioOnline-BD0E2BF97AA949AEA8BAEB0A4622533C

ResourceGroupName : rg-smartbank-lab
Location          : centralindia
ProvisioningState : Succeeded
Tags              :
ResourceId        : /subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-smartbank-lab

ResourceGroupName : rg-smartbank-test
Location          : centralindia
ProvisioningState : Succeeded
Tags              :
ResourceId        : /subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-smartbank-test

ResourceGroupName : cloud-shell-storage-centralindia
Location          : centralindia
ProvisioningState : Succeeded
Tags              :
ResourceId        : /subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/cloud-shell-storage-centralindia

ResourceGroupName : rg-smartbank-cli
Location          : centralindia
ProvisioningState : Succeeded
Tags              :
ResourceId        : /subscriptions/d780980c-d1c6-4cf2-9353-c5838b3e1627/resourceGroups/rg-smartbank-cli

ResourceGroupName : rg-cli-demo2
Location          : centralindia
ProvisioningState : Succeeded
```

5. ARM Template Deployment

Storage-template.json > [] resources > {} 0

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "resources": [
    {
      "type": "Microsoft.Storage/storageAccounts",
      "apiVersion": "2022-09-01",
      "name": "ibadmin123",
      "location": "centralindia",
      "sku": {
        "name": "Standard_LRS"
      },
      "kind": "StorageV2",
      "properties": {}
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell
bash

```
PAVILION@cheeky MINGW64 /d/VSC_AZURE
$ az deployment group create --resource-group rg-smartbank-lab --template-file Storage-template.json
            "centralindia"
          ],
          "properties": null,
          "resourceType": "storageAccounts",
          "zoneMappings": null
        }
      ]
    }
  ],
  "provisioningState": "Succeeded",
  "templateHash": "9472885733516312389",
  "templateLink": null,
  "timestamp": "2026-04-08T06:40:22.449777+00:00",
  "validatedResources": null,
  "validationLevel": null
},
"resourceGroup": "rg-smartbank-lab",
"tags": null,
"type": "Microsoft.Resources/deployments"
}
```